South Asia's Digital Revolution

UPI, India Stack, and the DPI Playbook for the Global South

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Executive Summary

South Asia's digital transformation — anchored by India's extraordinary Digital Public Infrastructure (DPI) architecture and spreading in varying forms to Bangladesh, Sri Lanka, Nepal, Pakistan, and Bhutan — is the most consequential development innovation in the region since the Green Revolution of the 1960s. India's "India Stack" — the layered architecture of Aadhaar (biometric identity), UPI (real-time payments), DigiLocker (digital documents), ONDC (open commerce), and Account Aggregator (consented data sharing) — has demonstrated that a developing economy can build population-scale digital infrastructure faster, cheaper, and with broader coverage than any private-sector-only model could achieve. India's UPI processed 131 billion transactions worth approximately $2.3 trillion in FY2025 — more digital payment transactions than the rest of the world's digital payment systems combined. The Aadhaar system has enrolled 1.38 billion individuals and processes 100+ million identity verifications daily.

At the G20 India Presidency summit in September 2023, Prime Minister Modi formally proposed India Stack as a global public good — offering to share the architecture, code, and implementation experience with developing nations for free. The "DPI Global Playbook" is now India's most consequential soft-power offering: a blueprint for how low-income countries can build population-scale financial inclusion, identity, and digital commerce infrastructure without dependence on Big Tech platforms (Google Pay, Amazon, Facebook's Libra/Diem) that extract data and extract fees, or on traditional financial institutions that have historically excluded the poor. South Asia's digital experience — particularly the DPI model — is reshaping how the Global South thinks about digital governance, financial inclusion, and the role of the state in digital infrastructure.

Part I — India Stack: The Architecture of Digital Inclusion

Four Layers, One Ecosystem

India Stack's architecture is conceptually elegant: four open-standard API layers, each building on the previous, creating a cumulative infrastructure that private companies can build products on top of without controlling the underlying infrastructure:

Layer 1 — Presence-less Layer: Aadhaar biometric authentication. Any service that needs to verify identity can call the Aadhaar API with a fingerprint or iris scan and receive a yes/no authentication response. This eliminates physical presence requirements for services — a bank account can be opened from a rural village, a SIM card provisioned, a government benefit claimed, using only Aadhaar-authenticated mobile authentication.

Layer 2 — Paperless Layer: DigiLocker (digital document storage and sharing — 6+ billion documents stored for 250 million users as of 2026; mark sheets, driving licences, vehicle registration, insurance policies, degree certificates all accessible via mobile app and shareable via QR code with government or private services).

Layer 3 — Cashless Layer: UPI — real-time, interoperable, zero-cost for merchants, accessible via feature phones (UPI 123PAY for 200 million feature phone users), and integrated with every bank account in India. 131 billion transactions FY2025.

Layer 4 — Consent Layer: Account Aggregator framework — enables individuals to share their financial data (bank statements, mutual fund holdings, insurance policies, tax records) with authorised financial institutions in machine-readable format, with explicit consent via mobile app, without physical paperwork. Enables credit underwriting for 300 million+ previously "thin file" borrowers who lacked traditional credit bureau records.

The genius of India Stack's design: each layer is an open API standard, not a government monopoly. PhonePe, Google Pay, Amazon Pay, and Paytm all built on UPI without paying licensing fees — but also without capturing UPI's infrastructure. The government owns the rails; private companies build the trains. This is the model that Big Tech platforms in other markets were never willing to enable — hence the zero-fee, interoperable architecture that is qualitatively different from WeChat Pay (closed ecosystem) or Visa/Mastercard (private toll-takers on payment rails).

Part II — Digital Financial Inclusion: From Exclusion to 1.38 Billion Identities

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The Jan Dhan-Aadhaar-Mobile (JAM) Trinity

India's financial inclusion revolution rests on the JAM Trinity: Jan Dhan Yojana (zero-balance bank accounts for the unbanked), Aadhaar (biometric identity linked to the account), and Mobile (smartphone or feature phone UPI access). The combination — launched as a deliberate government infrastructure programme in 2014-2015 — has been the most effective financial inclusion programme in world history by scale. 520 million Jan Dhan accounts have been opened; approximately 67% of adult Indians now have access to formal banking services (versus 35% in 2011). Government benefit transfers (LPG subsidies, MNREGA wages, pension payments, COVID relief payments) that previously leaked 30-40% to corruption intermediaries now flow directly into Jan Dhan accounts verified by Aadhaar — a savings of approximately $40 billion annually in benefit leakage by government estimates.

The pandemic stress test: COVID-19's emergency relief transfers were distributed to 200+ million vulnerable Indians within 10 days of program announcement — direct bank transfers to Jan Dhan accounts verified by Aadhaar, with zero physical infrastructure requirements. Compare this to the US, where COVID stimulus checks took 3-6 weeks for paper cheque delivery to unbanked households, and to UK, where emergency furlough payment infrastructure required weeks of bank detail verification. India's DPI had already solved this problem before the pandemic created it.

Credit for the Previously Unbanked

India Stack's Account Aggregator layer is enabling a new generation of credit underwriting that is potentially transformative for South Asia's 400 million+ "thin file" borrowers — individuals who have income, assets, and repayment capacity but lack the formal credit history that traditional banks require. By combining cash flow analysis (GST invoices for merchants, bank statement patterns, mobile recharge history, utility payment records), alternative data (satellite crop yield estimates for farmers, app usage patterns), and machine learning underwriting models, fintech companies (MoneyTap, KreditBee, BankBazaar, Lendingkart, Capital Float) are extending credit to borrowers who previously could access only informal moneylenders at 40-120% annual interest rates.

Part III — Bangladesh's Digital Journey: bKash and the Mobile Money Model

bKash: The Bangladesh Payment Revolution

Bangladesh's digital payments success is built on a different model than India's — not government-led DPI but private-sector mobile money. bKash — launched in 2011 as a subsidiary of BRAC Bank, the development finance institution, with investment from Money in Motion and Bill & Melinda Gates Foundation — became Bangladesh's dominant mobile money platform and one of the world's largest mobile financial services providers. bKash has approximately 80 million registered accounts (nearly half Bangladesh's adult population), processes approximately $25 billion in transactions annually, and enabled the RMG sector's shift to digital wage payment (eliminating cash salary queues that created both security and wage theft risks in garment factories).

bKash's success demonstrates that the mobile money model — pioneered by M-Pesa in Kenya — can scale even in a dense, low-income urban environment like Dhaka. Bangladesh has now extended bKash's infrastructure into government benefit transfers, utility payments, and cross-border remittance receipt. The government's Digital Bangladesh initiative (launched 2009, upgraded to "Smart Bangladesh" 2022) has built a network of Union Digital Centres providing digital services to rural populations who lack smartphones.

BFIN and the National Digital Architecture

Bangladesh's National Payment Switch (NPS) — the interoperability infrastructure connecting bKash, Nagad, and bank mobile apps to a common settlement system — provides the national digital payments layer analogous to UPI in India. Nagad (launched 2019 as a partnership between Bangladesh Post Office and Dutch-Bangla Bank) has grown rapidly as a government-linked alternative to bKash, with 75+ million users. The competitive two-platform market has driven fee reduction (both platforms now offer zero-fee person-to-person transfers) and innovation that benefits end users.

Part IV — The DPI Playbook: India's Global Export

India's DPI Diplomacy

India's presidency of the G20 in 2023 produced the G20 Framework for Systems of Digital Public Infrastructure — a formal international framework endorsing DPI as a development model and committing G20 members to support developing countries in building DPI. India's Digital Public Goods Alliance participation, its India-UN Tech Sprint, and its bilateral DPI cooperation agreements (UPI corridors with Singapore, UAE, France, Bhutan, Nepal, Sri Lanka, and Mauritius) collectively constitute an unprecedented digital diplomacy campaign.

The geopolitical dimension: DPI competes with the Big Tech model (where digital infrastructure is owned and controlled by Amazon, Google, Meta, and Apple — primarily US companies — with their data architecture centred on surveillance capitalism and network lock-in) and the Chinese model (Alipay/WeChat Pay as closed ecosystems that export Chinese digital standards alongside Chinese investment). India's DPI model — open standards, government infrastructure, private application layer — offers developing countries a third path that preserves national data sovereignty and avoids either US Big Tech or Chinese digital dependency. For countries in Africa, Southeast Asia, and Latin America that are being courted simultaneously by US tech companies and Chinese digital investors, India's DPI model provides a genuinely different alternative architecture with a 25-year implementation track record.

Conclusion: The Digital Dividend

South Asia's digital revolution — led by India's DPI architecture and Bangladesh's mobile money ecosystem — represents the most significant leap in financial inclusion, government service delivery, and economic productivity in the region's post-independence history. The JAM Trinity has extended banking access to half a billion previously excluded Indians in a decade; UPI processes more digital payment transactions than any other nation; and India Stack's open API architecture is being studied and adopted by governments from Africa to Southeast Asia as the template for 21st-century digital governance.

The opportunity ahead: as the Account Aggregator layer matures, it can enable a credit revolution for 400 million+ thin-file borrowers; as ONDC (Open Network for Digital Commerce) scales, it can disintermediate Amazon and Flipkart by providing small merchants direct digital market access; as the DPI export programme accelerates, India's digital standards can become the foundation of the Global South's digital infrastructure — a soft power dividend of extraordinary scale. South Asia's digital decade has begun. The question is whether the governance quality, privacy protection, and inclusive design that made UPI a global benchmark can be sustained as the DPI ecosystem scales into financial services, healthcare, and education.

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All data from NPCI, UIDAI, RBI, bKash, Bangladesh Bank, G20 DPI Framework, and cited sources.